Agricultural ICT

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Agricultural ICT (A13165)

Short Title:	Agricultural ICT		
Faculty	Science and Computing		
Department	Science		
Cluster	IT		
Author	ADUNPHY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The aim of this module is to provide students with the knowledge and understanding of a number of information and communication technologies that are quickly becoming crucial to the development of the agricultural sector. These are technologies that enable the gathering, storage and presentation of key farm data, such as sensor technologies, databases, web development and information systems.

Programmes

`COMP-0536`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	4 / 7 / M

Indicative Content

- Data: obtaining, analysing and displaying data; Data Protection Act; agricultural information systems, data governance
- Databases: agri-data management; database theory & principles; database design; database creation and manipulation
- Web development: develop an interactive website for the agricultural area & critique other agri-oriented websites
- Precision agriculture: sensors, potential applications of precision agriculture, its benefits & drawbacks
- Technology: basic components of a computer system and their function

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the importance of agricultural data and its governance by obtaining, analysing and displaying same.
2. Develop a database application for an agri-environment.
3. Develop a website for an agri-environment.
4. Determine the key concepts of precision agriculture and evaluate associated products. `.
5. Appraise different technical components.

Learning and Teaching Methods

- Practical demonstrations.
- Exercises.
- Practice.
- Lectures.
- Projects.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Final Project	40%	2
Assignment	25%	3
In-Class Assessment	35%	1,4,5

Assessment Criteria

<40%: Lacks technical ability/skill and understanding to carry out practical tasks.
40%-49%: Shows some technical ability/skill in carrying out and practical tasks.
50%-59%: Logically and competently carries out practical tasks.
60%-69%: Demonstrates sound knowledge of the subject matter.
70%-100%: Demonstrates imaginative approach, excellent technical ability, awareness and aptitude in carrying out and reporting on practical tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Essential Material

- "Moodle page for Ag ICT." moodle.wit.ie. https://moodle.wit.ie

Supplementary Material

- Heege, H.J. _Precision in Crop Farming; Site Specific Concepts and Sensing Methods; Applications and Results_. Dordrecht: Springer Science+Business, 2013.

Requested Resources

- Room Type: Computer Lab